

Build A PC — Cheat Sheet

Last-minute recall reference. For the reasoning behind each row, see LESSON.pdf or the interactive tour.

CONCEPT	RULE
Socket match	Must match motherboard exactly — check firmware support for newer CPUs too
RAM generation	DDR4/DDR5 set by the motherboard, not the CPU — physically different notches
QVL	The motherboard's own tested-compatible RAM list — check before buying
PSU sizing	Headroom above minimum, for transient spikes — not just the label wattage
Most forgotten cable	CPU power (EPS, 4/8-pin) — separate from the 24-pin
Test-boot order	Outside the case first — isolates component vs. assembly problems
XMP	Not on by default — enable in UEFI to hit RAM's actual rated speed
Thermal paste	Pea-sized amount — more is not better
No-POST order	Power → RAM reseal → CPU power cable → standoffs → debug LEDs
PSU quality	Protection circuitry (OCP/OVP/SCP) matters more than wattage alone
Thermal throttling	Automatic downclocking under heat — presents as slow, not broken